Lecture 24

STAT 109: Introductory Biostatistics

Lecture 24: One numerical variable: center, graphs, and the one-sample *t*-test

Learning objectives

- Compute **mean** and **median** by hand and with the **tidyverse** (pipe to `summarize()`).
- Graph one numerical variable with **ggplot2**: **histogram** and **dot plot**.
- Recognize the **one-sample *t*-statistic** and use `t.test()` for a **one-sample *t*-test** and **confidence interval** for the population mean.

Review: one numerical variable

A **numerical** variable takes values that are counts or measurements where it is meaningful to sort them in order and to compute distance between any two of them. We usually report center and spread; this lecture focuses on center and on graphs for the whole distribution.

Mean (average). For values $x_1, \dots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The capital Greek letter Sigma (Σ) denotes **summation**. The expression $\sum_{i=1}^{n} x_i$ means: add $x_1 + x_2 + \cdots + x_n$, including every observation of x, indexed by i from 1 to n. Multiplying that total by $\frac{1}{n}$ rescales it: you are taking the combined amount (the total sum) and expressing it as if it belonged to a single unit, "per person," "per observation," or "per employee" in a payroll setting.

How to read the mean. Think: total of all values, spread evenly across n slots. If everyone's contribution were replaced by the same number while keeping the same total, that common number would be the mean.

Visual (balance point). On a **dot plot** with one dot per observation, the mean is the **balance point** (center of mass) along the horizontal axis: equal weight at each dot, and the axis would balance at $\bar{x}$. On a **histogram**, you get the same idea if you treat each bin's count (or properly scaled density) as sitting at the **bin center**, that is a standard approximation; the exact mean always comes from the original values, not from the drawn bins.

Median. Sort the values from smallest to largest. If n is odd, the median is the middle value. If n is even, it is the average of the two middle values. In the usual case with no ties sitting exactly on the split, the median is the value such that half of the observations are below it and half are above it. (If many values tie at the median, the "50% / 50%" story is still right in spirit but the counts on each side need a careful definition, we keep the ordered-list rule for computing it.)

How to read the median. The median is a positional measure: it marks the center of the ordered list, not the balance of the total of the numbers. It is not pulled in dollars-and-cents by a few huge values in the same way the mean is.

Example: salaries. Suppose you list every employee's annual salary. The **median** answers a "typical employee" question: half of the staff earn less than or equal to that amount and half earn more than or equal to it. The median is a useful picture of how pay feels in the middle of the workforce. The **mean** answers a "budget" question: if the firm paid the same total in salaries but split it evenly across every employee, each would get the mean. So the mean is the per-employee share of the fixed payroll total; it is what you need if you want to think about resizing the organization ("what if we had twice as many people with the same total wage bill?") or track aggregate cost per head.

Which should you report? There is no single correct choice for every problem. If the goal is to describe the feeling of a typical standing, the median is often easier to interpret. If the goal is to tie summaries to a total, a rate per unit, or planning with a fixed sum divided among n units, the mean is a better choice. Report what matches your question and when in doubt, both plus a graph can be clearer than either alone.

Examples by Hand

1. Data: 7, 8, 9, 10, 10, 11, 12, 13 .

- Mean: $\bar{x} = (7 + 8 + 9 + 10 + 10 + 11 + 12 + 13)/8 = 80/8 = 10$.
- Median: notice that the 8 values are already sorted from smallest to bigest and the two middle values are both 10, so the median is 10.

Here the mean and median match.

2. Data: 2, 3, 4, 5, 6, 7, 100 (seven values). One value (100) is far from the rest.

- Median: with seven values already sorted, the median is the fourth in the list so the median is 5.
- Mean: the sum is $2 + 3 + 4 + 5 + 6 + 7 + 100 = 127$, so $\bar{x} = 127/7 \approx 18.1$.

The mean is pulled toward the large observation; the median stays with the middle of the bulk of the data. For describing a "typical" case here, the median (5) is the point at which about half the values are less than it and half are greater than it. The mean reflects the total (127) spread across seven slots, including the one very large contribution and is the "balance point" of all seven values.

Mean and median in the tidyverse (pipe)

Syntax note: In **dplyr**, `summarize()` and `summarise()` are the same function (American vs. British spelling). For pipes, base R's native `|>` (R 4.1+) works like the **magrittr** pipe `%>%` from the **tidyverse**: both pass the left-hand side into the first argument of the function on the right. This lecture uses `|>` and `summarize()`; use whichever spelling and pipe you prefer, they behave the same here.

The pipe `|>` sends the data on its left into the first argument of the function on its right. We often pipe a data frame (e.g. a dataset with examples in rows and variables in columns) into `summarize()` to compute summaries.

```r
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

scores <- tibble(
  score = c(7, 8, 9, 10, 10, 11, 12, 13)
)

scores |>
  summarize(
    n = n(),
    mean_score = mean(score),
    median_score = median(score)
  )
```

```
# A tibble: 1 x 3
```

```
      n mean_score median_score
  <int>      <dbl>        <dbl>
1     8         10           10
```

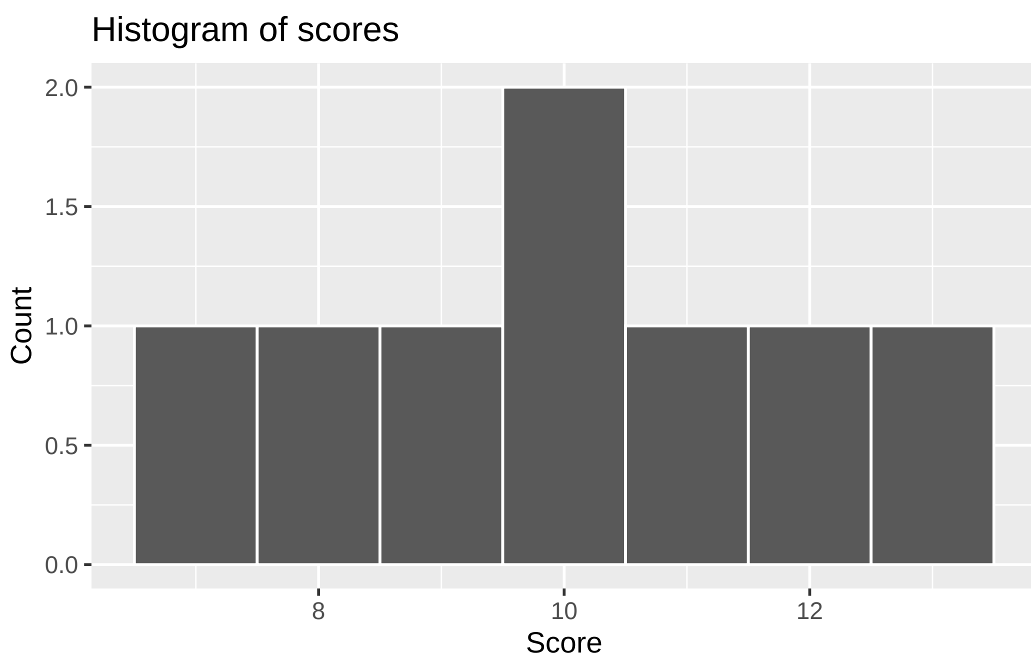
Here `n()` counts rows inside `summarize()` (one row per student, say).

Histogram and dot plot (ggplot2)

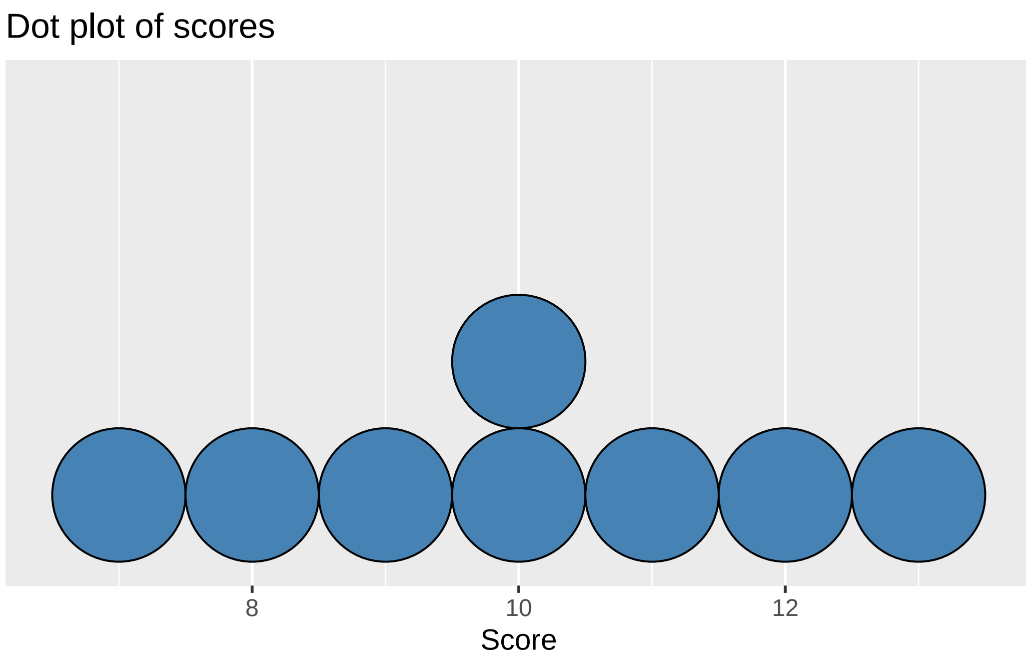
Histogram, split the axis into bins and show counts (or densities) in each bin.

Dot plot, stack dots along the axis so you see each value (or rounded value) and repeats clearly.

```
ggplot(scores, aes(x = score)) +
  geom_histogram(binwidth = 1, boundary = 6.5, color = "white") +
  labs(
    title = "Histogram of scores",
    x = "Score",
    y = "Count"
  )
```




```
ggplot(scores, aes(x = score)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "steelblue") +
  labs(
    title = "Dot plot of scores",
    x = "Score",
    y = NULL
  ) +
  scale_y_continuous(NULL, breaks = NULL) # hide stacking axis
```



Adjust `binwidth` so the plot matches how you think about the scale (e.g. group by every 1 unit, 10 units, 100 units…) or leave this input out to let R use a default setting.

One-sample t-statistic and `t.test()`

Setting. We have a **simple random sample** of size n from a population with unknown mean μ. We want to test whether μ equals a hypothezed value μ_0 or to estimate reasonable values for μ via a **confidence interval**.

Definition: t-statistic (one sample). With sample mean $\bar{x}$ and sample SD s,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The denominator is the **standard error** of the mean. Under $H_0 : \mu = \mu_0$, and when the usual conditions hold (random sample and either n is large (>30) or the population is roughly symmetric), t is compared to a **Student *t* distribution** with $\mathrm{df} = n - 1$. The exact shape depends on sample size and is not calculable by hand. In modern times, we use software for p-values and intervals, not tables by hand.

`t.test()` for one sample: pass the numeric vector, the null mean `mu`, and optionally `alternative` (`"two.sided"`, `"greater"`, `"less"`). The default output includes a test and a 95% **confidence interval** for μ (two-sided alternative).

```
t.test(scores$score, mu = 9, alternative = "two.sided")
```

```

	One Sample t-test

data:  scores$score
t = 1.4142, df = 7, p-value = 0.2002
alternative hypothesis: true mean is not equal to 9
95 percent confidence interval:
  8.327958 11.672042
sample estimates:
mean of x
```

```
10
```

Interpretation. The **confidence interval** is the set of estimates for u that we believe contain the true value of μ with \$95\\%\$ confidence. The **p-value** summarizes evidence against $H_0 : \mu = \mu_0$ in favor of the stated H_a: reject the null H_0 in favor of H_a if the p-value is less than the significance level α (often taken to be 0.05 but remember you should pick this value based on the seriousness of a Type I error).

Takeaway

For a single **numerical** variable, **mean** and **median** describe center; **ggplot2** gives histograms and dot plots of the full distribution. For inference about one population mean, the **one-sample *t*-test** and interval come from `t.test()` on the sample vector.